Homework 1

Part A

1 $z = W^T x + b$

$= [1, 3] \begin{bmatrix} 2 \\ -1 \end{bmatrix} - 2$

$= (2-3) - 2$

$= -3$

2 $z = wx + b$

$= 4 \cdot -0.5 + 1$

$= -1$

$y = \sigma(z)$

$= \frac{1}{1+e^{-1}}$

$= \frac{1}{1+e}$

$= 0.26$

3. $z_1 = W_1 x + b_1$

$= \begin{bmatrix} 1 & -1 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$= \begin{bmatrix} -1 \\ 2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$= \begin{bmatrix} -1 \\ 3 \end{bmatrix}$

$h = ReLU(z_1)$

$= \begin{bmatrix} \max(0,-1) \\ \max(0,3) \end{bmatrix}$

$= \begin{bmatrix} 0 \\ 3 \end{bmatrix}$

$y = W_2 h_1 + b_2$

$= [2 \ -1] \begin{bmatrix} 0 \\ 3 \end{bmatrix} + 0$

$= -3$

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4. $L = \frac{1}{2}(\hat{y} - y)^2$

$= \frac{1}{2}(-1-2)^2$

$= \frac{1}{2}9$

$= \frac{9}{2}$

5. $L = \frac{1}{2}(\hat{y} - y)^2$

$\frac{\partial L}{\partial \hat{y}} = \hat{y} - y \qquad \frac{\partial L}{\partial \hat{y}} = -(\hat{y} - y)$

6. $\hat{y} = w^T x + b$

$= [0\ 1] \begin{bmatrix} 1 \\ -2 \end{bmatrix} + 1$

$= -1$

$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial w_1}$

$= (\hat{y} - y)\, x_1$

$= -1 \cdot \begin{bmatrix} 1 \\ -2 \end{bmatrix}$

$= \begin{bmatrix} -1 \\ 2 \end{bmatrix}$

$\frac{\partial L}{\partial w_2} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial w_2}$

$= (\hat{y} - y) \cdot x_2$

$= -1 \cdot -2$

$= 2$

$\frac{\partial L}{\partial b} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial b}$

$= -1 \cdot 1$

$= -1$

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7. $\eta = 0.2$

$$w_{new} = w - \eta \begin{bmatrix} \frac{\partial L}{\partial w_1} \\ \frac{\partial L}{\partial w_2} \end{bmatrix}$$

$$= w - 0.2 \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 1 \end{bmatrix} - 0.2 \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 0.2 \\ -0.4 \end{bmatrix}$$

$$= \begin{bmatrix} 0.2 \\ 0.6 \end{bmatrix}$$

$$b_{new} = b - \eta \frac{\partial L}{\partial b}$$

$$= 1 - 0.2 \cdot -1$$

$$= 1.2$$

8. $z = wx + b$

$= 2 \cdot -1 - 1$

$= -3$

$a = RELU(z)$

$= \max(0, -3)$

$= 0$

$$\frac{\partial L}{\partial w} = \frac{\partial L}{\partial a} \frac{\partial a}{\partial z} \cdot \frac{\partial z}{\partial w}$$

$= (a - y) \cdot 0 \cdot -1$

$= 0$

, since the derivative of RELU is undefined at 0, I will just set it to 0 for convenience

$$\frac{\partial L}{\partial b} = \frac{\partial L}{\partial a} \frac{\partial a}{\partial z} \frac{\partial z}{\partial b}$$

$= (a - y) \cdot 0 \cdot -1$

$= 0$

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9.

Step 1:

$\hat{y} = 0.$, $y = 1$

$\frac{\partial L}{\partial w} = (\hat{y} - y) \cdot x$ $\qquad$ $\frac{\partial L}{\partial b} = -1$

$= -1 \cdot 2$

$= -2$

$w_1 = w_0 - \eta \frac{\partial L}{\partial w}$ $\qquad$ $b_1 = b_0 - \eta \frac{\partial L}{\partial b}$

$= 0 - 0.1 \cdot 2$ $\qquad$ $= 0 - 0.1 \cdot -1$

$= -0.2$ $\qquad$ $= 0.1$

10 Step 2:

$\hat{y} = -0.2 \cdot 2 + 1$

$= 0.6$

$\frac{\partial L}{\partial w} = (0.6 - 1) \cdot 2$ $\qquad$ $\frac{\partial L}{\partial b} = -0.4$

$= -0.8$

$w_2 = w_1 - \eta \frac{\partial L}{\partial w}$ $\qquad$ $b_2 = b_1 - \eta \frac{\partial L}{\partial b}$

$= -0.2 - 0.1 \cdot -0.8$ $\qquad$ $= 0.1 - 0.1 \cdot -0.4$

$= -0.2 + 0.08$ $\qquad$ $= 0.1 + 0.04$

$= -0.12$ $\qquad$ $= 0.14$

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10. $S = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$

$$\text{softmax}(S) = \begin{bmatrix} \frac{1}{1+e^1+e^2} \\ \frac{e^1}{1+e^1+e^2} \\ \frac{e^2}{1+e^1+e^2} \end{bmatrix}$$

Part B:

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In forword pass, the input goes through linear combination and activations to give an estimate the real value.

In back propagation, we update the parameters and bias, by using gradient desent.

12.

The neural network will not be able to predict non-linear relationship. besides, our input value can become really large in forward propagation and our partial derivatives will become really large in gradient descent. Our weight and bias value can become really unstable.

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$$\text{softmax}(s) = \begin{bmatrix} \frac{e^0}{e^0 + e^1 + e^2} \\ \frac{e^1}{e^0 + e^1 + e^2} \\ \frac{e^2}{e^0 + e^1 + e^2} \end{bmatrix}$$

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13.

If learning rate is too large, we can update our weight and bias too fast, and we may skip over local minimum and maximum point.

If our learning rate is too small, we can make no update to our weight and bias and we may take forever to reach local minimum or maximum

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vanishing gradients means that gradients become really small as we propagate back through layers. When gradient becomes really small, we will not be able to update weights and bias, especially when there are a lot of layers in the network. Thus, it can be a problem for training deep networks

15. ReLU doesn't shrink input value and keeps output non-negative. Thus, it can alleviate the vanishing gradient problem.

In contrast, sigmoid function can suffer from vanishing gradient problem greatly